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Editors of *Mathematics of Computation*
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Dear Editors,

I submit the manuscript “**Frequency Saturation and Certified Computation of Relaxation Functionals**” for consideration in *Mathematics of Computation*.

The paper concerns the computation of linear functionals of a positive distribution of relaxation times from impedance measurements, the Stieltjes-type inverse problem $Z(\omega) = R + i\omega L + \int (1 + i\omega\tau)^{-1} d\mu(\tau)$ with unknown nonnegative resistance and inductance. It determines how much frequency information such a computation requires, and it gives a finite numerical method, with a complete termination proof, that computes the entire data-consistent range of the functional with certified numerical error.

The main results are the following. (1) Exactly $K + 2$ distinct frequencies are necessary and sufficient for uniform noiseless identification of references with at most K atoms among all finite positive measures, and $K + 1$ when the calibration parameters are known. (2) At this threshold a fixed design attains the worst-case accuracy order of the entire positive-frequency response; the sharp exponent is determined by local target regularity and cluster multiplicity, uniformly through colliding atoms and vanishing masses. Positive moment-matched spectra that stay close at every frequency give the lower bound, including against target-aware adaptive queries, and calibration-annihilating polynomial reproduction gives the upper bound. (3) An exact-arithmetic refinement algorithm encloses both endpoints of the continuous feasible range with any prescribed numerical excess; finite optimization gaps, target approximation, and continuum correction enter separate error budgets, and the certificates yield an a posteriori absolute–relative stopping rule. (4) The same modulus gives matching expected acquisition costs for prescribed terminal width and confidence under a Gaussian variance–effort law, against adaptive frequency, precision, and stopping; a class-tuned expanding design that invokes the certified computation attains this order and terminates at every finite positive spectrum.

The work joins the analysis of a restricted information class to an executable, accuracy-controlled inverse computation, and so addresses approximation of positive measures, information complexity, and verified numerical methods. The numerical section reports exact full-frequency certificates, equal-effort design comparisons, certified cost bounds, and executed stopping paths; a reproducibility archive with exact inputs, verification code, and certificates accompanies the submission.

The manuscript is original, unpublished, and not under consideration elsewhere. I declare no conflicts of interest. I suggest Professor Markus Bachmayr as a handling editor, given his expertise in approximation, adaptive methods, and inverse problems.

Sincerely,

Haotian Zhong